Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-2)

BIOCHEMISTRY

Subject Code : BP-203T

M.Code : 74969

Date of Examination : 19-12-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION - A

1. Answer the following :

a. Define free energy and redox potential.

b. What are energy-rich compounds? Give two examples.

c. Write two glycogen storage diseases.

d. Differentiate between gluconeogenesis and glycolysis.

e. Define uncouplers of oxidative phosphorylation.

f. Mention two disorders of lipid metabolism.

g. Write the role of the urea cycle in nitrogen metabolism.

h. Give two examples of neurotransmitters derived from amino acids.

i. What is semi-conservative replication of DNA?

j. Define isoenzymes with one example.

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SECTION - B

2. Describe the citric acid cycle with energetics and significance.

3. Discuss the *de novo* synthesis of fatty acids and their regulation.

4. Explain enzyme kinetics with the Michaelis-Menten and Lineweaver-Burk plots.

SECTION - C

5. Explain glycogen metabolism and its regulation.

6. Discuss oxidative phosphorylation and substrate-level phosphorylation.

7. Write a note on ketone bodies and their role in ketoacidosis.

8. Explain the catabolism of tyrosine and related metabolic disorders.

9. Discuss the genetic code and inhibitors of protein synthesis.

10. Explain the regulation of enzyme activity (allosteric enzymes with examples).

11. Write a note on the HMP shunt and its significance.

12. Explain the electron transport chain with its mechanism.

13. Write a note on transcription and RNA synthesis.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.